Voting for a consensus with veto rights

Open proposals, full ballots and a blocking quota

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October 2026

Research seminar

Outline

The question
What no procedure can achieve
What research says about veto rights
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The procedure: Condorcet with a blocking quota
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The question

A group must choose one option, by consensus

A site, a design, a budget allocation, a policy. Three hypotheses, set by the use case:

H1 (open agenda) anyone may propose as many options as they wish;

H2 (full ballots) everyone judges *all* the options proposed;

H3 (veto rights) voters can veto.

Two requirements:

R1 the option chosen is the best compromise;

R2 the group never gets stuck: the veto must not be too stringent.

R1 and R2 pull against each other. Unanimity protects every minority, but one objector blocks everything (the *liberum veto*). Majority always decides, but can impose on a large minority.

The design question: how much veto power, held by whom, over what, and how to aggregate the rest of the ballot.

The answer in one slide

QVC(θ): Condorcet with a blocking quota, $\theta = 1/3$ by default

1. Each voter ranks every option and marks the unacceptable ones.
2. Options vetoed by at least θn voters are excluded.
3. If that excludes every option, the least-vetoed ones are kept.
4. The Schulze winner among the options left is elected.

- Always decides, in one round; the status quo has no default advantage.
- The winner is acceptable to more than $1 - \theta$ of the voters whenever some option is.
- Among acceptable options, the majority decides; adding variants of an option changes nothing.
- The price is strategic vetoing: a 10% coalition can manipulate 11–27% of structured elections, against 5–7% under Schulze.

What no procedure can achieve

Impossibility results fix the margins of the choice

- **Manipulation is always possible.** With three options or more, every onto, non-dictatorial rule is manipulable Gibbard (1973), Satterthwaite (1975). The question is how often, by whom, at what cost.
- **The majority principle has a cost.** Every Condorcet-consistent rule violates participation with four options or more: the no-show paradox Moulin (1988); tight at 12 voters, Brandt, Geist & Peters (2017).
- **Majorities can cycle, but rarely do in structured electorates.**
  - Impartial culture, large n: a Condorcet winner exists with probability 91.2% for three options, 74.9% for five Guilbaud (1952), Gehrlein & Fishburn (1976).
  - Our spatial and polarised electorates: at least 96% even with twelve options; real ballots agree Regenwetter et al. (2006), Tideman & Plassmann (2012).

  A procedure must handle cycles, but they are the exception.

Proposition 1: majority and minority veto conflict

Example E1, 100 voters, status quo C:

Voters	Ranking	Veto
51	$A \succ B \succ C$	C
49	$B \succ C \succ A$	A

- A is the Condorcet winner: 51–49 against both.
- B is everyone's first or second choice; 49% rank A last and veto it.

Proposition 1

For $m \geq 3$ and $0 < \theta < 1/2$, some profiles have a Condorcet winner vetoed by more than θn voters while another option is acceptable to all. No rule can both elect the Condorcet winner and respect a θ-veto there.

Choosing θ is choosing between A (majority) and B (compromise). Majority power and veto power trade off across rules Kondratev & Nesterov (2020).

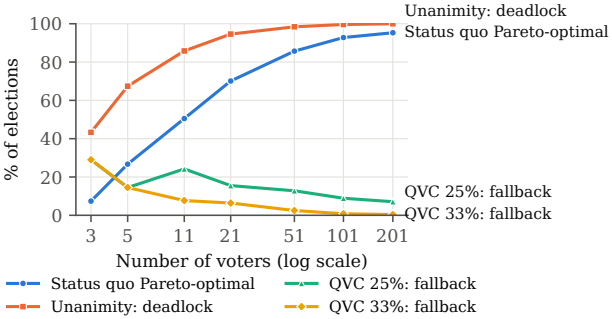
Stability and deadlock are two faces of the veto

- A stable outcome exists for every profile iff m is below the game's Nakamura number Nakamura (1979): majority has 3, hence cycles; a veto player has ∞, hence stability.
- Spatial thresholds: a supermajority above $d/(d+1)$ in d dimensions Greenberg (1979); 64% under concave densities Caplin & Nalebuff (1988). Below them, agenda control can lead the vote almost anywhere McKelvey (1976).
- But the stable outcome a veto guarantees is the status quo.

Proposition 2 (deadlock under unanimity)

Voters prefer options closer to their ideal point $p_i \in \mathbb{R}^d$. (i) If $s \in \text{conv}(p_1, \ldots, p_n)$, no option is preferred to s by every voter: unanimity with the status quo as default keeps s, whatever the proposals. (ii) If the p_i are i.i.d. from a distribution whose support contains a neighbourhood of s, the probability of (i) tends to 1 as $n \to \infty$.

Deadlock in numbers



Two dimensions, seven proposals and the status quo. Unanimity keeps the status quo in

- 43% of elections with 3 voters,
- 86% with 11,
- 99.6% with 101.

QVC falls back on the least-vetoed options instead, and still decides.

Buchanan & Tullock (1962): decision costs rise with the quota. Proposition 2 says they become infinite in a large assembly under individual veto.

What research says about veto rights

Unanimity against majority with a veto

- **Information aggregation.** Majority rule with a separate veto beats unanimity in large committees: a doubter can vote against without blocking Bouton, Llorente-Saguer & Malherbe (2018); laboratory, 2017. Separate blocking the unacceptable from approving a choice.
- **Bargaining.** Unanimity brings delay, conflict and surplus captured by holdouts Miettinen & Vanberg (2025), Kagel, Sung & Winter (2010), Agranov & Tergiman (2019).
- **Veto players.** More veto players, more policy stability, i.e. gridlock Tsebelis (2002); an endogenous status quo adds inertia Dziuda & Loeper (2016); extra rounds strengthen the vetoer Nunnari (2021).

Two implications for the design:

1. the status quo must not be a privileged default, or every veto becomes a vote for inertia;
2. no sequential agenda: the order of votes can be manipulated McKelvey (1976), Barberà & Gerber (2017). Under H2, everyone judges all options at once.

Compromise rules with vetoes, and consensus in practice

Rules

- Voting by veto Mueller (1978)
- Proportional veto core: x% of the voters veto x% of the options Moulin (1981, 1982); polynomial Ianovski & Kondratev (2021)
- Fallback bargaining Brams & Kilgour (2001)
- Majoritarian compromise Sertel & Yılmaz (1999)
- Shortlisting, veto-rank Barberà & Coelho (2022), Anbarci (1993)

Several fail as compromises once the result is known Cailloux, Napolitano & Sanver (2022). None was designed for an open agenda.

Practice

- IETF rough consensus: objections addressed, not necessarily accommodated RFC 7282
- Tacit consent: the formal veto is socially constrained Krick (2017)
- *Liberum veto*: paralysis Roháč (2008)
- EU Council: a blocking minority needs at least four members TEU, art. 16
- Debian: ranks all options with "None of the above"; quorum and majority against the default; Schulze. The closest existing procedure, but the default stays privileged.

What the literature supports

Claim	Evidence
Full unanimity is rarely optimal for open choices: delay, status-quo bias, holdouts	Strong
Majority with a separate veto beats unanimity for information aggregation	Strong
No rule is strategy-proof; the majority principle and minority vetoes conflict	Theorem
One simultaneous ballot removes agenda power; clone independence removes the gain from flooding	Strong (theory)
Condorcet winners exist in most structured electorates	Strong
Fallback, shortlisting and veto-rank rules are second-best compromise devices	Moderate
One or two rounds suffice in structured proposal settings	Moderate
In practice, a formal individual veto is socially constrained	Weak to moderate

The note, Table 2: adapted and extended from the Consensus review of October 2026.

An open agenda calls for independence of clones

Flooding the agenda

Clones are variants of one proposal: no voter ranks another option between them, and every voter vetoes all or none Tideman (1987). Under H1, anyone can create them.

- Plurality: clones split their supporters' votes and lose. Borda: they push the others down and win.
- Independent of clones: Schulze, Ranked Pairs, Split Cycle. Not: Copeland, Minimax, Kemeny–Young.
- Veto-based rules are exposed, because m enters their definition: in the proportional veto core, each clone the majority adds costs the minority a veto.

Proposition 3 (flooding defeats proportional veto)

Neither the proportional veto core nor fallback bargaining is independent of clones.

In E1, the 49% block A. In **E2** the majority adds two variants of A: the minority's vetoes no longer cover them all, and A enters the core and wins.

The rules compared

Rule	Condorcet winner	Independent of clones	Monotonic	Uses vetoes	Never stuck
Plurality	×	×	✓	×	✓
Instant-runoff (IRV)	×	✓	×	×	✓
Borda	×	×	✓	×	✓
Copeland, Minimax	✓	×	✓	×	✓
Schulze, Ranked Pairs, Split Cycle	✓	✓	✓	×	✓
Approval	×	✓[a]	✓	✓[b]	✓
Fallback bargaining	×	×	✓	×	✓
Unanimity, status quo default	×	✓	✓	✓	×
Schulze within the PVC	×	×	—	✓[c]	✓
QVC(θ)	×[d]	✓	✓[e]	✓	✓

×: fails in some profiles; —: not established; PVC: proportional veto core. [a]When clones are approved alike. [b]Non-approval acts as a veto mark but blocks nothing. [c]Veto power read from the bottom of the rankings. [d]Elects the Condorcet winner of the admissible options (Proposition 6). [e]For changes in the rankings that leave the veto marks unchanged. The note, Table 3.

The procedure: Condorcet with a blocking quota

Six design principles

P1 One simultaneous ballot on the whole agenda: no one controls the order of votes.

P2 Ranking and blocking are separate acts: a ranking plus an acceptability line.

P3 Vetoes are held by coalitions, a share θ of the voters, not by individuals.

P4 The status quo is an option like the others: ranked, vetoed, elected, never the default.

P5 The aggregation is clone-independent: flooding the agenda gains nothing.

P6 A decision is guaranteed: if every option is blocked, keep the least-vetoed.

P1–P2: Bouton et al. (2018), McKelvey (1976). P3: Propositions 1–2. P4: Tsebelis (2002), Dziuda & Loeper (2016). P5: Proposition 3. P6: R2.

QVC(θ)

Ballot: a ranking $\succ_i$ with an acceptability line; V_i, the options below it. Counts: $v(x) = |\{i : x \in V_i\}|$ and $\mathrm{N}(x, y) = |\{i : x \succ_i y\}|$.

Admissible options, θ-compromise

$A_\theta = \{x : v(x) < \theta n\}$ if this set is not empty, else $\arg\min_x v(x)$. The *θ-compromise* is the Condorcet winner of A_θ, when it exists.

1. Count the vetoes; form A_θ (never empty).
2. Elect the Schulze winner of A_θ. Ties: fewest vetoes, then Schulze on the tied options, then an order drawn before the vote.
3. Publish v, A_θ and the pairwise matrix N, so that anyone can check the count.

Lexicographic reading of R1: first acceptable to all but a minority smaller than θ, then majority. $O(nm^2 + m^3)$ operations; Ranked Pairs could replace Schulze.

What QVC(θ) guarantees

Prop. 4 **Decisive, with no default advantage.** Exactly one option, in one round, for every profile; every option, the status quo included, is treated alike; the order of proposal plays no role.

Prop. 5 **Acceptability.** If some option is vetoed by fewer than θn voters, so is the winner. No rule elects an option vetoed by a blocking minority less often.

Prop. 6 **Majority within the admissible options.** The winner is in the Smith set of A_θ and not Pareto-dominated; the Condorcet winner of the whole agenda wins whenever fewer than θn voters veto it.

Prop. 7 **Independence of clones**, up to the random order that breaks exact ties: a clone inherits the vetoes of the original, and Schulze is clone-independent.

Proofs: the note, Appendix A. The code checks Propositions 4–7 on random profiles.

The price: strategic blocking

Proposition 8

Any coalition of at least θn voters can make any option inadmissible by vetoing it. For every $\theta \leq 1/2$, there are profiles in which, whatever the winner w, at least θn voters prefer another option to w, and can block w although they find it acceptable.

- The strategic face of Proposition 1: the power to block the unacceptable is also the power to block the acceptable.
- An option that no coalition of size θn wants to replace exists for every profile only if $\theta > (m-1)/m$, i.e. only if the veto is weaker than majority Greenberg (1979).
- The proportional veto core avoids the problem, but fails independence of clones (Proposition 3), which H1 makes essential.

The counting rule cannot remove this; the procedure around the vote must contain it.

Four worked examples

Winners in the worked examples

Example	Plur.	Borda	Schul.	Fallb.	PVC	Unan.	Q 25%	Q 33%	Q 50%
E1 (compromise)	A	B	A	B	B	B	B	B	A
E2 (flooding)	A	A	A	A	A	B	B	B	A
E3 (cycle)	A	A	A	B	A	C (stuck)	B (fb)	A	A
E4 (status quo)	X	X	X	Y	X	S (stuck)	X	X	X

- **E1**: the choice of θ decides between A (majority) and B (compromise).
- **E2**, flooding: Borda, fallback and the proportional veto core switch to A; QVC does not move.
- **E3**, a cycle: with $\theta = 1/4$ every option is blocked and QVC falls back; with $\theta = 1/3$ the weakest option is excluded. Unanimity is stuck.
- **E4**, the status-quo trap: one voter vetoes both proposals; unanimity keeps the status quo against 99 voters.

Simulation evidence

Design of the Monte Carlo study

- 101 voters, 8 options: 7 open proposals and the status quo. Proposals are made by voters near their own position, so larger blocs propose more and near-duplicates arise, as under H1.
- Three electorates: **spatial** (ideal points on a plane), **polarised** (two blocs, 55% and 45%), **impartial** (random rankings, a worst case).
- Vetoes from a personal tolerance fixed before the menu is known: a veto marks the unacceptable, not the below-average.
- 21 rules, 2,000 elections per electorate. Measures:
  - Condorcet efficiency; utility and worst-off (10th percentile) efficiency, scaled from 0 for the average option to 100 for the best Merrill (1984);
  - contested: the winner is vetoed by at least a third; deadlock: the status quo stays;
  - manipulability by full coalitions and by coalitions of at most 10% of the voters.

Every number in the talk comes from the code (`simulations/python/`), with checks against known results.

Results: the polarised electorate

Rule	Condorcet eff.	Utility eff.	Worst-off eff.	Contested	Deadlock
Plurality	22.4	34.4	13.3	62.1	0.0
Borda	71.2	94.5	79.1	15.2	0.0
Approval	72.2	95.5	95.0	5.9	0.0
Fallback bargaining	53.7	87.7	79.9	18.6	0.0
Unanimity, status quo default	16.2	8.5	30.7	57.4	97.9
Schulze	100.0	97.4	85.2	10.1	0.0
Proportional veto core + Schulze	90.8	96.2	86.7	10.2	0.0
QVC(1/4)	83.2	96.8	92.8	5.9	0.0
QVC(1/3)	94.3	97.7	88.4	5.9	0.0

- Condorcet methods lead on welfare; hard vetoes deadlock.
- QVC(1/3) cuts contested outcomes from 10.1% to 5.9%, the minimum any rule can reach (Prop. 5), and raises worst-off efficiency from 85.2 to 88.4, at no cost in total utility. The cost: Condorcet efficiency 94.3%.

2,000 elections, 101 voters, 8 options; in %. The note, §7, which also gives the spatial and impartial electorates.

Flooding and manipulation

Three clones of a losing option change the winner in

- up to 54% of elections: proportional veto core + Schulze;
- up to 36%: fallback bargaining;
- up to 24%: Borda (teaming);
- 0%: Schulze; 0%: QVC(1/3).

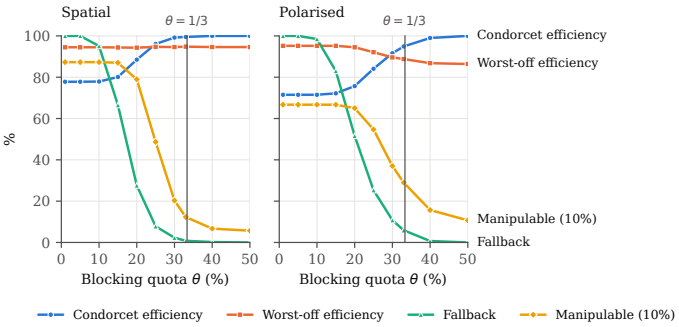
Cloning the winner defeats it under plurality in 98% of spatial elections (vote splitting).

Coalitions of at most 10% of the voters can manipulate (spatial–polarised)

- Schulze: 5–7%;
- QVC(1/3): 11–27%; QVC(1/4): 52–57%;
- instant-runoff: 37–45%; approval: 85–69%.

Full coalitions manipulate most rules in most elections, QVC with $\theta < 1/2$ in nearly all (Prop. 8). Three strategies only: lower bounds.

Choosing the quota



- Low quotas: the fallback decides most elections, the veto is only a negative vote, and the result is easily manipulated. High quotas: the veto never binds and QVC is Schulze.
- $\theta = 1/3$: fallback rare (5.9% of polarised elections), Condorcet efficiency at least 94.3%, most of the worst-off gain kept. Close to the EU's 35% and to the 36% of the 64% rule; $\theta = 1/4$ for a group that values consensus more.

Safeguards, limits and open questions

Containing strategic vetoes

- **Named and reasoned vetoes.** A veto is public and states a reason, as an objection must in rough consensus RFC 7282; responsibility changes votes Breitmoser & Valasek (2024).
- **One revision round, only if the fallback was used.** Nominations reopen briefly for compromise proposals; the second ballot is final. One or two rounds suffice; more strengthen vetoers Gersbach & Valkanova (2024), Nunnari (2021).
- **Veto budgets across decisions, not within one** storable votes, Casella (2005). A budget within one decision would break independence of clones.
- **No lower quota than needed**: manipulability falls steeply as θ rises.

The status quo can still be protected on purpose, Debian-style, for costly or irreversible decisions.

How firm, and what is open

How firm

- Theorems: manipulation is unavoidable; majority and minority veto conflict; unanimity deadlocks a large assembly.
- Broad literature: Condorcet methods lead on welfare; an open agenda needs clone independence.
- Judgement: $\theta = 1/3$. The shape of the trade-off is robust across cultures; the numbers are not.

Limits: one decision; fixed preferences, no deliberation; equal weights; stylised electorates.

Open questions

- A clone-proof proportional veto? Clone independence, veto power proportional to size and a non-empty core: compatible, or not?
- Fast one- or two-round procedures with hard vetoes in several dimensions.
- A strategic comparison of compromise rules under common assumptions.
- Do named vetoes reduce strategic blocking? A laboratory test.
- Real ballots with acceptability lines: Debian, PrefLib.

Conclusion

- The research consensus is negative about unanimity and positive about vetoes that are separate from approval and limited to blocking the unacceptable.
- Social choice adds: no escape from manipulation; majority and minority veto must be traded; an open agenda requires independence of clones.
- One design follows: QVC(1/3). One ballot; exclude what a blocking third rejects; never stuck; Schulze among the rest.
- It never deadlocks, protects the worst-off, ignores flooding, and costs no total welfare. Its weakness is that of any veto: strategic blocking, to be contained by procedure.

Note, code and data: `github.com/FabriceBM/Vote_Condorcet`

Backup: the Schulze method

On a set S, the link from x to y has strength $d(x,y) = \mathsf{N}(x,y)$ if $\mathsf{N}(x,y) > \mathsf{N}(y,x)$, else 0. The strength $p(x,y)$ of the strongest path is the best, over paths from x to y, of the weakest link. Floyd–Warshall for widest paths:

$$p(x,y) \leftarrow \max\big(p(x,y),\ \min(p(x,z), p(z,y))\big) \quad \text{for } z \in S, \text{ then all } x, y,$$

from $p = d$, in $O(|S|^3)$. Winners: the x with $p(x,y) \geq p(y,x)$ for every y; there is always one.

Condorcet- and Smith-consistent, clone-independent, monotonic, Pareto Schulze (2011). Used by Debian.

Backup: testing the proportional veto core by max-flow

$W_i(x)$: the options voter i ranks at or below x, x included.

Lemma

x is in the proportional veto core iff $\left|\bigcup_{i \in T} W_i(x)\right| \geq m|T|/n$ for every coalition T; equivalently, the maximum flow equals mn in the network

$$\text{source} \xrightarrow{m} i \xrightarrow{\infty} y \in W_i(x) \xrightarrow{n} \text{sink}.$$

A cut keeping the voters of T on the source side must keep $\bigcup_T W_i(x)$ there; its capacity is $m(n-t) + n\left|\bigcup_T W_i(x)\right|$. One max-flow per option, on $n+m+2$ nodes; checked against the definition by enumeration on small profiles. Polynomial algorithm: Ianovski & Kondratev (2021).

Backup: proof of Proposition 2(i)

Suppose y is strictly preferred to s by every voter: $\|y - p_i\| < \|s - p_i\|$ for all i. Expanding the squares,

$$(y - s) \cdot p_i > \tfrac{1}{2}\big(\|y\|^2 - \|s\|^2\big) \quad \text{for all } i,$$

so every ideal point lies in an open half-space H, which is convex and contains $\mathrm{conv}(p_1, \ldots, p_n)$. But $s \notin H$:

$$(y - s) \cdot s - \tfrac{1}{2}\big(\|y\|^2 - \|s\|^2\big) = -\tfrac{1}{2}\|y - s\|^2 < 0.$$

Hence $s \notin \mathrm{conv}(p_1, \ldots, p_n)$. Under unanimity with the status quo as default, a proposal passes only if every voter prefers it to s, so none passes. □

(ii): a small simplex around s inside the support; each of its $d + 1$ vertex balls receives an ideal point with probability $\to 1$.

Backup: putting the procedure to work

1. **Fix the parameters in advance**: the quota ($\lceil n/3 \rceil$ voters by default), whether the status quo is protected (by default not), the nomination period, the revision round.
2. **Nomination**: anyone proposes any number of options; the status quo, or "none of the above", is on the ballot.
3. **Ballot**: rank or grade every option, ties allowed; mark the unacceptable ones; a veto is named and reasoned. Grade form: excellent, good, fair, poor, unacceptable.
4. **Count** with QVC; publish the veto counts, the admissible options, the pairwise matrix and the winner; anyone can check it with the reference code.
5. **Revision round**, only if every option was vetoed by the quota; the second result is final.

Small groups: the quota is a number of voters; below five voters, two vetoes, so that no single member blocks alone.

Backup: the simulated electorates

- Proposals: a voter drawn at random proposes an option at their ideal point plus Gaussian noise (s.d. 0.25); the status quo is drawn independently. Utility: minus the Euclidean distance.
- Spatial: ideal points $\sim \mathcal{N}(0, I_2)$. Polarised: blocs of probability 0.55 and 0.45 centred at $(\pm 1.2, 0)$, s.d. 0.6. Impartial: utilities i.i.d. uniform.
- Vetoes: beyond a personal tolerance, $\rho_i \sim U(1.5, 2.5)$ in distance (spatial cultures), fixed before the menu, so clones inherit the original's vetoes.
- Efficiency of winner w for a score S: $(S(w) - \bar{S})/(\max_x S(x) - \bar{S})$, S the sum of utilities or their 10th percentile across voters.
- Manipulation: the voters who prefer y to the winner (all of them, or the first ten) try three joint ballots; a success counts. Shares are lower bounds.
- Each election has its own seed: the tables are identical from run to run.

The note, Appendix C; `cultures.py`, `voting.py`, `experiments.py`.